

Separating a Growing Neutrino Mass from Dynamical Dark Energy: the Scale-Dependent Growth Signature of an Evolving MOND Scale

Carl P. Zimmerman · Briar Creek Tech · 2026-06-28 A companion/forecast note to “A Growing Neutrino Mass from an Evolving MOND Scale” (DOI 10.5281/zenodo.20977421).

Abstract

The de Sitter–Unruh modified-inertia proposal sets the MOND acceleration scale by the dark-energy density, $a_0 = cH_\Lambda/Z = (c/2)\sqrt{(G\rho_{DE})} \approx 9.36 \times 10^{-11} \text{ m s}^{-2}$, so a_0 evolves as $a_0(z)/a_0(0) = \sqrt{(\rho_{DE}(z)/\rho_{DE}(0))}$ on the DESI-preferred evolving-dark-energy branch. The companion note showed that *if* the lightest neutrino is a swampland-tower state, its mass is then **locked to $a_0(z)$ with no free amplitude**, freezing by recombination so that the CMB imprints a $\sim 25\text{--}45\%$ lighter neutrino than today’s — a right-signed, right-magnitude reading of the DESI “negative effective neutrino mass” anomaly. The honest objection that note left open is **degeneracy**: a constant neutrino mass with *generic* dynamical dark energy also pushes the inferred Σm_ν below the oscillation floor. Here we identify and quantify the one observable that breaks the degeneracy. A *growing* neutrino mass changes the free-streaming fraction $f_\nu(z)$, suppressing structure growth **only below the neutrino free-streaming scale $k_{fs}(z)$ — a scale-dependent** step in the matter power spectrum. Dynamical dark energy changes the expansion $H(z)$ and is **scale-independent**. The scale-dependence is therefore a separator no equation-of-state mirage can reproduce. A real CAMB calculation confirms the step ($\approx 0.65\%$ at the normal-ordering floor, $\approx 1.4\%$ at $\Sigma = 0.10 \text{ eV}$), and a focused Euclid Fisher gives a detectability of $\approx 1.4\sigma$ at the floor rising to $\approx 3.9\sigma$ at $\Sigma = 0.10 \text{ eV}$ — i.e. the test’s power is **hostage to the absolute neutrino mass**, which JUNO and KATRIN pin independently this decade. The prediction is **robust to the DESI DR1→DR2 update** (the locked mass ratio moves by < 0.05). It remains **conditional** (on $\alpha \approx \lambda$ and $\nu = \text{tower}$), **founded-not-derived** (the absolute mass is free), and **dies if $w \rightarrow -1$** . It is **not** a theory of everything.

1. The claim, and the degeneracy it must beat

The companion note’s content reduces to one falsifiable statement: the same evolving $a_0(z) = \sqrt{\rho_{DE}}$ that the framework already predicts forces, *if the neutrino is a tower state*, a neutrino mass that grows from past to present, **locked to $\rho_{DE}^{1/2}$** with no free amplitude. Propagating the public DESI DR1 $w_0 w_a$ posterior gives $m_\nu(\text{CMB})/m_\nu(\text{today}) \approx 0.54\text{--}0.72$; the mass freezes through recombination, so a constant-mass CMB+BAO fit under-counts the late-time mass and infers $\Sigma m_\nu^{\text{eff}} \approx 0.032\text{--}0.042 \text{ eV}$ — in the DESI low/negative band.

The objection: the mainstream reading of the same anomaly is *generic dynamical dark energy* mimicking a low Σm_ν . Both shift the **amplitude** of the inferred mass below the floor. On amplitude alone they are degenerate. A real test must find an observable where the two part ways.

2. The degeneracy-break: scale-dependence

They part ways in the **shape**, not the amplitude.

- A **growing neutrino mass** raises $f_v(z) = \Omega_v(z)/\Omega_m(z)$ toward the present. Massive neutrinos free-stream out of potential wells below the free-streaming scale, so they suppress the growth of structure **only for $k > k_{fs}(z)$** ($k_{fs} \approx 0.007\text{--}0.016\text{ h Mpc}^{-1}$ for the masses of interest). The signature is a **scale-dependent step** in $P(k)$: suppressed small-scale power, intact large-scale power.
- **Dynamical dark energy** alters the background expansion $H(z)$, which rescales the growth factor **uniformly in k** . It is **scale-independent**.

Therefore the scale-dependence of the late-time growth deficit is a separator that no $w(z)$ can fake. This is a *qualitative*, physical fork (free-streaming vs background), not a numerical coincidence.

CAMB confirmation. Replacing the linear $\Delta P/P \approx -8 f_v$ rule of thumb with real CAMB transfer functions (reviews/growing_nu_camb_fisher.py), comparing the present-day power for the true (heavier, today's) mass against the CMB-anchored (lighter) constant-mass fit:

Σm_ν today	small-scale suppression	($k > 0.2$)	large-scale ($k < 0.01$)	scale-dependent step
0.059 eV (NO floor)	1.09%		0.44%	0.65%
0.10 eV (above floor)	2.00%		0.61%	1.39%

The step is real and lives at the free-streaming scale — exactly where dynamical dark energy leaves no feature.

3. Detectability: hostage to the absolute mass

A focused Euclid Fisher for the **scale-dependent** part of the signal (subtracting its k -mean to marginalize any scale-independent amplitude or dark-energy contribution) gives:

$\approx 1.4\sigma$ at the normal-ordering floor ($\Sigma=0.059\text{ eV}$); $\approx 3.9\sigma$ at $\Sigma=0.10\text{ eV}$.

The detectability scales with f_v , i.e. with the **absolute** neutrino mass. This is the actionable result: at the oscillation floor the test is marginal, but even a modestly higher mass makes it a real detection — and the absolute mass is being pinned **independently** by JUNO (first data 2026) and KATRIN this decade. The growth-tomography test does not stand alone; its power is set by a number other experiments are already measuring.

4. Robustness to the DESI DR1→DR2 update

The companion note used DESI DR1 (2024). Recomputing the locked mass ratio across both the DR1 best-fits and the published DESI DR2 (2025) central $w_0 w_a$ values (reviews/growing_nu_dr2_sensitivity.py):

$m_\nu(\text{CMB})/m_\nu(\text{today})$: **DR1 0.56–0.71 → DR2 0.57–0.74** — a shift of <0.05 , well inside the supernova-sample spread.

The prediction is robust to the data update; DR2’s main effect is to *tighten* the evidence for evolving dark energy ($2.8\text{--}4.2\sigma$), the condition on which the whole prediction is hostage. (The DR2 cobaya chains were not accessible from our environment; DR2 values are published centrals, to be verified against arXiv:2503.14738.)

5. Honest scope

- **Founded-not-derived.** The absolute m_ν is free; the framework forces the *ratio* $m_\nu(z)/m_\nu(0)$ (via $\Delta\phi$ fixed by the measured $w(z)$) and the *scale* $E_{dS} = \rho_{DE}^{1/4} = 2.24 \text{ meV}$ — not the mass itself.
- **Conditional** on $\alpha \approx \lambda$ (a swampland conjecture, not a theorem) and on the lightest neutrino being a tower state.
- **Approximate.** The CAMB step uses a two-constant-mass bracket for the true $m(z)$ interpolation (it captures the endpoint signature); the Fisher uses rough single-tracer volume/number-density and does not marginalize galaxy bias or nonlinear scales. A publication-grade forecast needs the full Boltzmann $m(z)$ evolution and a multi-tracer Fisher.
- **Dies if $w \rightarrow -1$.** No roll $\rightarrow$ no field excursion $\rightarrow$ no mass evolution $\rightarrow$ no signal (not a falsification, just a null).
- **Neutrino/dark sector only — not a theory of everything.** The charged-lepton and quark masses are untouched and walled.

The contribution of this note is narrow and concrete: it converts the companion paper’s qualitative “testable by a joint fit” into a **specific, physical, dark-energy-unfakeable observable** (the free-streaming step) with a **computed detectability** that sharpens with the absolute neutrino mass.

References

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Reproducible: `reviews/{growing_nu_fsigma8_forecast, growing_nu_camb_fisher, growing_nu_dr2_sensitivity}.py` (exit 0). Footing $a_0 = 9.36 \times 10^{-11} \text{ m s}^{-2}$, $Z = \sqrt{(32\pi/3)}$. Companion: DOI 10.5281/zenodo.20977421.